

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
FIRST YEAR (Software Engineering)
SPRING SEMESTER EXAMINATIONS 2025

Time : 3 Hours

Batch 2024

Dated : 02-JUN-25

Max Marks : 60

Software Engineering - SE-207

Instructions: Attempt All Questions

Q#1.

- a. **Explain** the scope of a small application that analyzes each course offered by a university and reports the average grade obtained in the course for a given term. Your explanation should summarize the project description, clearly describe the problem to be solved, and outline the suggested solution.

CLO4
[07]

- b. You read "The Manifesto for Agile Software Development". Can you think of a situation in which one or more of the four "values" could get a software team into trouble Show your understanding by **describing** why these values could lead to difficulties in that context.

[03]

Q#2.

Design a booking system for a hotel. **Draw** a use case diagram and a sequence diagram for the process of booking a room through the system.

CLO2

- a. **Use Case Diagram:** Illustrate the main actors involved in the system, such as **Guest** and **Hotel Staff**. Identify key use cases related to the booking process, including **Browse Rooms**, **Select Dates**, **Make Reservation**, **Make Payment**, **Modify Reservation**, and **Cancel Reservation**. Show the relationships between actors and use cases, indicating which use cases each actor can initiate or participate in.

[06]

- b. **Sequence Diagram:** Draw a sequence diagram specifically for the **Make Reservation** process. Identify the actors involved, sequence the interactions between these actors and the system components during the booking process, and include the messages exchanged, the sequence of actions, and any conditional logic or decisions (e.g., room availability check, payment confirmation).

[05]

Q#3.

- a. **List** the key management activities involved in software project management. For each activity, briefly explain its purpose and how it contributes to the success of a software project.

CLO1
[05]

- b. Garvin identified eight dimensions of product quality that help evaluate and compare products, including software. Select five of Garvin's quality dimensions, **define** each dimension in general terms and highlight why focusing on just one or two dimensions (instead of a balanced approach) might lead to poor software quality.

[05]

P.T.O

Q#4.

CLO4

a. You are leading a team tasked with developing a secure online banking system for a large financial institution. Given the critical importance of reliability, security, and regulatory compliance, which software process model would you recommend: the Waterfall model or the Incremental model? **Analyze** the system requirements and discuss how each model addresses the project's challenges and constraints. Provide examples of specific scenarios where each model would excel in ensuring a successful project outcome.

[06]

b. **Describe** the XP concepts of refactoring and pair programming in your own words.

[04]

Q#5.

CLO2

a. **Demonstrate** the relationship between the concept of information hiding as an attribute of effective modularity and the concept of module independence.

[03]

b. **Apply** your understanding of requirements engineering to develop a set of 3 functional requirements and 3 non-functional requirements for a library management system. Ensure that each requirement reflects the system's operational needs and user expectations.

[06]

Q#6.

CLO1

a. **Define** the incremental testing strategy and describe how it can be applied when testing a software system. Provide examples to illustrate your explanation.

[06]

b. **What** do you think happens when requirement validation uncovers an error? Who is involved in correcting the error?

[04]